Complications Of Local Anesthesia

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Complications of local anesthesia can be divided into two areas. There are local complications and systemic complications. Local complications are the result of the mechanics of the injection or the properties of the anesthetic drug on the local environment. Systemic complications are those general complications occurring as a result of the drug used or a local problem that can lead to systemic sequels.

Local complications

1- Pain on injection:

Pain on injection can be prevented through careful adherence to the basic rules of atraumatic injection.

Causes:

1- Palatal injection always hurt.
2- Dull needle from multiple injections with the same needle.
3- Rapid deposition of the anesthetic solution.
4- Needle with barbs; impaling the needle on bone may produce a "fishhook barb'' that produces pain as the needle withdrawn from the tissue.

Prevention:

1- Carefully adhere to proper technique of injection, both anatomically and psychologically.
2- Use sharp needles.
3- Use topical anesthetic prior to injection.
4- Practice slow injection of solutions.
5- Be certain that the temperature of the solution is correct; cold solutions stored in refrigerator usually, produce pain on injection.

2- Burning on injection:

Burning during deposition of the local anesthesia is not uncommon.

Causes:

1- PH of the solution; local anesthetics are prepared to be slightly alkaline and cause no pain. Those containing vasoconstrictor are more acidic PH. This usually causes mild burning sensation.
2- Rapid injection of local anesthetic solution especially in the denser more adherent tissue of the palate.
3- Contamination of the local anesthetic cartilages with sterilizing solution as result of diffusion of this solution into the cartilage.

4- Temperature of the solution even if it's warm.

The problem is transient in nature and is an indication of tissue irritation. The condition rapidly disappears when the local anesthetic action develops with no residual sensitivity after action termination of the lost sensation. However, there is greater opportunity of the tissue damage to develop with subsequent postanesthatic trismus, edema, or paresthesia.

Prevention:

1- Slow injection; the ideal rate is 1ml/min; don't exceed 1.8 ml in 1 min.

2- Proper care and handling of the local anesthetic cartilage; it is stored at room temperature in suitable container without alcohol or any other sterilizing solutions.

3- Persistent anesthesia; paresthesia:

It is a rare, disturbing complication of local anesthesia. Tissue injury in the form of biting or thermal or chemical insult can occur without a patient's awareness.

Causes:

1. Trauma to the nerve or the nerve sheath.
2. Injection of contaminated local anesthetic cartilage by alcohol or sterilizing solutions near the nerve with resultant irritation and edema that increase pressure on the region. This may cause paresthesia.
3. Hemorrhage into or around the neural sheath also may increase pressure on the nerve and produce paresthesia.

Management:

Most paresthesia resolves in 8 weeks without treatment and will be permanent only if there is severe nerve damage.

1. Reassure the patient:
   a. The dentist must talk to the patient and explain that it's not uncommon complication after injection.
   b. Record the incident in the dental chart.
2. Examine the patient:

   Determine the degree of paresthesia and inform the patient that paresthesia may persist at least 2 months or may be prolong to one year.
3. Reschedule the Patient:

   Reschedule the patient for examination every 2 months as long as sensory deficit persists. Should sensory deficit still be evidence one year after that, consult an oral surgeon or neurologist to mange.

1- Trismus:

Trismus is a motor disturbance of the trigeminal nerve especially spasm of the masticatory muscles with difficulty in opening the mouth. Although post injection pain is the most common local anesthetic complication, trismus can become one of the more chronic and complicated problem to manage.

Causes:

1- Trauma to muscles or blood vessels in the infratemporal space is the most common cause of trismus following the dental injections.
2- Contaminated dental cartilage by diffusion of alcohol or any sterilizing agent with the resultant irritation to the muscle is potential cause of trismus.
3- Hemorrhage (large volume of blood) produce tissue irritation and possibly muscle dysfunction as blood is slow resorbed.
4- A low grade infection.
5- Multiple needle penetrations.
6- Overly large amount of local anesthetic solution deposited in restricted area.

In most cases patient will report pain and difficulty in opening the mouth the day after the dental appointments in which posterior superior alveolar or inferior alveolar nerve blocks were administered. The advice is to use sharp, sterile, disposable needles, cleanse the site of injection with an antiseptic solution prior to needle penetration, avoid repeat injections and multiple insertions of the needle, and use minimal effective volumes of local anesthetic solutions.

Management:

1- Heat therapy: Placing moist heat with towels to the affected area about 20 min every hour.
2- Analgesics: Aspirin is usually adequate in management of pain associated with trismus. Codeine (30-60 mg every 3 hours) may be prescribed if the discomfort is more intense.
3- Muscle relaxants: Diazepam (about 10 mg every 12 hours).
4- Advice the patient to initiate physiotherapy for 5 min every 3-4 hours by opening and closing the mouth as well as lateral excursions and by intake of chewing gum(sugarless).
5- Record the incident, finding, treatment in the patient's dental chart.
6- Avoid any further dental treatment in the involved region till symptoms resolves and patient is more comfortable.
7- Seven full days Antibiotics is required if the pain and dysfunction continued beyond 48 hours due to possibility of infection.

8- Refer the patient to oral surgeon if no improvement within 2-3 days without antibiotics or 5-7 days with antibiotics or severe limited mouth opening.
9- TMJ involvement is quite rare in the 1st 4-6 weeks following injection.
10- Surgical intervention may be indicated in some instance.

2- Hematoma:

The inadvertent nicking of a blood vessel, either artery or vein during an injection of local anesthetics is the cause of hematoma formation. Nicking of a vein may or may not cause hematoma. While nicking an artery usually produce hematoma that increases rapidly in size. The density of the tissue surrounding the injured vessel will be a determining factor e.g. hematoma is rarely developed after palatal injection but usually following nicking of the blood vessel in posterior superior or inferior alveolar nerve block.

Management:

Immediate:

When swelling becomes evident, direct pressure should be applied to the site of bleeding for not less than 60 sec. against bone. For inferior alveolar nerve block, pressure is applied to the medial aspect of the mandibular ramus. For infraorbital nerve block, pressure in applied to the skin directly over the infraorbital foramen. In mental nerve block, pressure is applied directly over the mental foramen, either on the skin or on the mucous membrane. On the other hand, if the condition occurs during posterior superior alveolar nerve block, the resultant hematoma is the largest and most esthetically unappealing. Further, hematoma is not recognized till the swelling appears on the side of the face progressing inferiorly and anteriorly toward the lower anterior region of the cheek. Difficulty in applying pressure at the site of the bleeding in this region is another complicating factor. Digital pressure can be applied to the soft tissues in the mucobuccal fold as far as it can be tolerated by the patient without gagging. Cold applications may be of value.

Subsequent:

Avoid any additional dental therapy in hematoma region till the signs and symptoms relived. The patient should be informed about the case and possible sequel of skin discoloration and trismus.

6- Infection:

The major cause of infection is carrying surface bacteria from non-sterile mucous membrane into the deeper structures with the tip of the needle. Another causative factor is Injection into infected area.

Prevention:

1- Use disposable needles.
2- Recap the needle when not in use to avoid contamination through contact with non sterile surfaces.
3- Avoid multiple injections with the same needle.
4- Cleanse the diaphragm of the cartilage with sterile, disposable alcohol wipes.
5- Properly prepare the tissues prior to penetration; dry the tissue and apply topical antiseptic.

Management:

Low grade infection is difficult to diagnose. Trismus may be sign. Treat trismus. If no response in three days then place the patient on antibiotic. Use penicillin or appropriate antibiotic if allergic to penicillin for at least seven days. If no response, refer the patient to Oral and Maxillofacial surgeon.

6- Edema:

Edema is the swelling of tissues.

Causes:

1- Trauma during injection
2- Infection
3- Allergy; Angioedema is a common reaction to topical anesthesia in an allergic patient. Localized tissue swelling occurs due to vasodilatation secondary to histamine release.
4- Hemorrhage; effusion of blood into soft tissues.
5- Injection of irritating solution (alcohol or cold sterilizing solution – containing cartilage).

Complications:

Airway obstruction is a serious complication of tissue edema in maxillofacial region. Angioedema in allergic patient may compromises the airway. Edema of the tongue, pharynx, and larynx may develop a life-threatening situation that needs vigorous management.

Prevention:

Properly care for and handlings of the local anesthetic armamentarium, together with complete and adequate medical examination of the patient prior to drug administration.

Management:

1- Management is predicated to reduce the swelling as quickly as possible.
2- Edema due to traumatic injection or introduction of irritating solution usually has a minimal degree of swelling and resolved within 1-3 days.
3- It's necessarily to prescribe analgesics for pain due to edema.

4- Hematoma following hemorrhage will resolve within 7-14 days.
5- Edema produced by infection will not resolve spontaneously but may Progressively be more intense. If the signs of infection (pain, mandibular dysfunction, and edema) are present, antibiotic should be prescribed.
6- Edema produced by allergy is the most potentially life threatening. The degree of the edema and its location is highly significant. If the swelling develops in the buccal soft tissue and there is no airway obstruction, intra-muscular and oral antihistamine administration will manage the swelling. Medical consultation to an allergist to determine the precise cause of the edema is recommended.

7- When edema compromised breathing:
a) Epinephrine 0.3 mg IM or IV
b) Antihistamine IM or IV
c) Corticosteroid IM or IV
d) Through evaluation of the patient and medical assistance.
e) Patient positioned in supine position if unconscious.
f) Basic cardiac life support.
g) Preparation of cricothyrotomy if total airway obstruction develops.

8- Sloughing of tissues:

Prolonged ischemia resulting from the use of local anesthesia with vasoconstrictors may lead to number of unpleasant complications including epithelial desquamation and sterile abscess. It has been strongly recommended, when using vasoconstrictors for homeostasis, not to employ overly concentrated solutions.

N.B: the palatal tissues are virtually the only tissues in the oral cavity where this phenomenon might occur.

Management is usually symptomatic.

9- Self inflicted injury, Lip-Chewing:

Trauma of the lips and tongue of the anesthetized patient is frequently caused by the patient inadvertently biting or chewing these structures. Trauma occurs most frequently in children, and mentally handicapped children and adult. The primary cause is the use of long acting local anesthesia in patient undergoing shorter dental procedures. Selection of proper duration of local anesthetic action depends on the duration of the dental procedures. A cotton roll can be placed between the lips of the patient if they are still anesthetized at time of discharge. Warn the patient against eating while still anesthetized, against drinking hot fluids, and against biting on the lips and tongue to test for anesthesia.

10- Facial nerve paralysis:

The facial nerve is the 7th cranial verve which is a motor nerve to the muscle of facial expression. Occasionally it can anesthetized by the inadvertent deposition of local anesthetics into its vicinity, always occur when the solution introduce in the deep lobe of the parotid gland. Transient facial nerve paralysis is commonly caused by the introduction of local anesthetic solution into the capsule of the parotid gland, which is located at the posterior border of the ramus of the mandible, clothed by the medial pterygoid and masseter muscles. Directing of the needle toward or its inadvertent deflection in a posterior direction during an inferior alveolar nerve block may place the needle tip within the substance of the parotid gland and paralysis may result. Loss of facial expression muscle's function will last from one to several hours depending on the local anesthetic agent, volume injected, and its proximity to the facial nerve.

The primary problem is unilateral paralysis during this time with inability to use theses muscle normally (cosmetic appearance problem). No treatment is required except waiting till the action wears off.

The secondary problem is that the patient is unable to close the eye, winking and blinking become impossible to perform. Advice the patient to periodically close the upper eyelid manually to keep the cornea lubricated. Contact lenses should be removed until muscular movement returns.

11- Needle Breakage, fig (1):

Become rare due to the introduction of disposable needle, but still occur. Causes:

1- Sudden unexpected movement of the patient as the needle penetrates the muscle or contacts periosteum.
2- Needles of smaller gauge.
3- Needles that have previously bent.

No problem exists if the needle can be easily retrieved without surgical intervention. Needles that break off within tissues and can't be readily retrieved usually enclosed by scar tissue and rarely produce infection. Leaving the needle is better than performing traumatic surgical removal. Inform the patient that the statement "needle will migrate with the blood" is not true.

Prevention:

1- Use larger gauge needle for injection. 25-gauge needles are appropriate for nerve block injections.

2- Use long needle for nerve block injection. Don't insert the needle into tissue to the hub (the point at which the needle shaft meets the hub is the weakest point of the needle).

3- Don't redirect the needle once it is inserted into tissues.

Management:

When the needle breaks:

a- Be calm, don't panic.

b- Instruct the patient not to move; don't remove your hand from the patient's mouth and keep the mouth opened and place a bite block if possible.

c- If the fragment is protruding, remove it with cotton pliers or a small hemostat.

If the needle is lost & can't be readily retrieved:

a- Don't precede with incision or probing if the fragment invisible.

b- Calmly inform the patient and relieve his fears and apprehension.

c- Note the incident in the patient's records.

d- Refer the patient for oral surgeon consultation.



Fig (1): Broken needle

12- Post anesthetic intraoral lesions:

Patient might report painful ulceration of the mouth following 2 days of dental injection. Trauma to tissues by needle, cotton swab, or any other instrument reactivate latent form of the disease process that has been present in the tissue prior to the injection.
Recurrent aphthous stomatitis &/or herpes simplex can develop intraorally following local anesthetic injection or any traumatic insult.

Recurrent aphthous stomatitis is the most frequently observed intraorally in the movable gingival tissue (not attached to the bone) e.g. buccal vestibule. It is not a viral infection but it might be autoimmune process or form bacterial infection.
Herpes Simplex can develop intraorally but it is most commonly extra-orally on the fixed tissue.
Management is symptomatic and directed for pain relief and keeping the ulcerated area covered.

Systemic complications

1- Syncope:

Fainting or Syncope: It is due to temporary acute cerebral ischemia, usually associated with needle insertion, as a result of dilation of the splanchnic blood vessels and reduced cardiac output.
Signs &symptoms: 1. Pallor, 2. Cold, 3. Sweaty, 4. Dizzy, 5. Nausea, and 6. Loss of consciousness.

Management:

1- Stop dental procedure.
2- Place the patient in supine position with slight head down or elevate the legs to increase cerebral circulation or
3- Place the hands of the patient behind his head and bend him forward until his head is in between the knee. This maneuver will produce pressure on the splanchnic blood vessels to replace the blood that drained from the head
4- Reassure the patient. Aromatic spirit (ammonia) held under the nose of the patient for several breaths.

2- Toxicity:

Definition: It refers to the symptoms manifested as a result of over dosage or excessive administration of the solution.

Etiology: 1- Large dose of drug, 2- Accidental intravenous injection, 3- High concentration of a drug, 4- Unusually slow detoxification as a result of severe liver disease, 5- Slow elimination as a result of kidney trouble, 6- Use of highly toxic drugs whose margin of safety is narrow, 7- Injection of solution in highly vascular area without the addition of vasoconstrictor substance.

The clinical picture may include stimulation (e.g., excitement, disorientation, increased heart rate and respiration, tremors, and frank convulsions) if inhibitory neurons are affected initially. CNS depression may cause hypotension, respiratory depression, unconsciousness, and death. Death usually occurs due to respiratory depression and hypoxia and its subsequent effect on the cardiovascular system.

Prevention: 1- Pre-analgesic evaluation of the patient; 2- Use the weakest possible concentration, 3- Use vasoconstrictor whenever possible, 4- Use of the least possible volume, 5- Aspirate before injection, 6-Slow injection, 7- Monitor the patient carefully after injection.

Treatment: includes supportive measures. Excitement and convulsions may be controlled with 5 mg doses of diazepam or 2 mg doses of midazolam. Respiratory depressions require oxygen and possibly rescue breathing. Cardiovascular system (CVS) derangement, High plasma titers may depress the cardiovascular system directly. Blood pressure may fall because of arteriolar dilation, myocardial depression, and/or cardiac conduction disruption. Treatment includes patient positioning, IV fluids, and vasopressors. Cardiac asystole will require Cardiopulmonary resuscitation (CPR). Toxicity to vasoconstrictors is very rare due to low content about 0.02mg.

3- Allergy and anaphylactic reaction:

Definition: Allergy is a hypersensitive reaction that occurs through exposure to an antigen (Ag) such as a drug (as local anesthetic agent) which the patient has been previously exposed to it creating an Ag – Ab (antigen-antibody) reaction. Anaphylaxis is a severe form of allergic reactions occurs suddenly and ends fatally in which sudden violent loss of vasomotor tonus.

Clinically: Allergy may be manifested as: 1.Rashes, 2. Urticaria, 3. Fever, 4. Dermatitis, 5. Angioneurotic edema, 6. Bronchial asthma, and 7. Anaphylaxis (anaphylactic shock). Anaphylaxis, on the other hands, is presented with:

1) Drop in blood pressure, 2). Weak rapid pulse, 3) Inadequate respiration (decreased respiratory rate), and 4) Death.

Management: 1- Shock position and clear airway, 2- Oxygen/ establish IV line, 3- Give corticosteroid, adrenaline, and antihistaminic, 4- Monitor vital signs and start CPR if necessary.

4- Idiosyncrasy:

Definition: Abnormal reaction to a drug or group of drugs exhibited when only a small amount or a nontoxic dose of the drug has been used. This means that the susceptible patient reacts adversely to a volume or concentration of the drug that would not affect the typical patient. Idiosyncrasy is nontoxic – non allergic condition.

Clinically: 1. Pallor, 2. Tachycardia, 3. Hypotension, 4. Decreased Heart rate, and 5. Collapse.

Prevention: 1. Do not use any drug, which the patient gives you a history of previous reactions to it 2. Inject slowly and observe the patient closely during injection. Treatment is the same as the depression stage of toxic overdose. Barbiturates are ineffective in the treatment prevention of this condition.